PROJECT TITLE: - DIFFERENTIAL EXSPRESSION AND SURVIVAL ANALYSIS OF TP53 IN BREAST INVASIVE CARCINOMA USING TCGA DATA.

A Primary Transcriptomic Analysis Using The Cancer Genome Atlas (TCGA) and GTEx Cohorts

- **Target Cancer:** "BREAST INVASIVE CARCINOMA (TCGA Code: BRCA)"
- **Target Gene: TP53** (Tumor Protein 53 – often called the "Guardian of the Genome")
- **Research Hypothesis:** TP53 gene expression is significantly altered in breast tumor tissue compared to normal breast tissue, and elevated TP53 expression correlates with poorer patient survival.

1. EXECUTIVE SUMMARY & INTRODUCTION

The tumor suppressor gene *TP53* (Tumor Protein p53), often termed the "Guardian of the Genome," encodes a critical transcription factor regulating cell cycle arrest, DNA repair, and apoptosis. In human malignancies, *TP53* function is frequently disrupted. This study presents a transcriptomic analysis comparing *TP53* gene expression across primary breast adenocarcinoma tissue and non-cancerous solid tissue controls using RNA sequencing (RNA-seq) data retrieved from **The Cancer Genome Atlas (TCGA)** and integrated GTEx control samples.

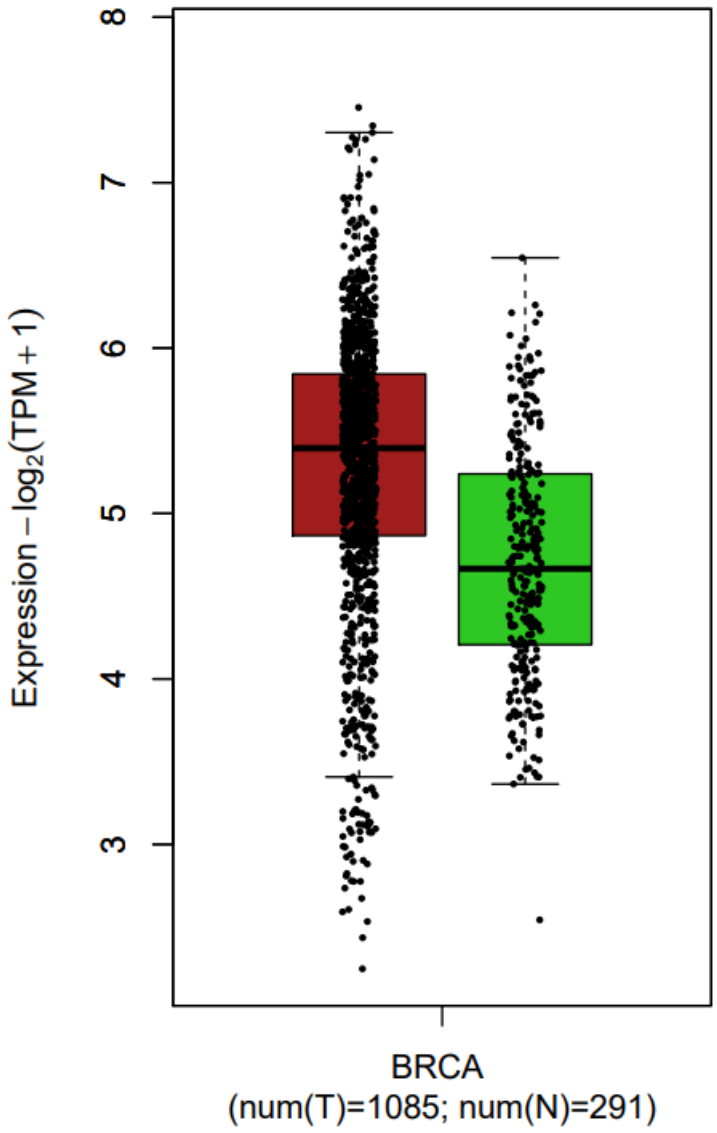
2. Research Methodology

Transcriptomic expression profiles were processed via the GEPIA2 (Gene Expression Profiling Interactive Analysis) web tool. Raw RNA-seq data were normalized in units of Transcripts Per Million (TPM) and transformed on a logarithmic scale:

$$Expression\ Value = \log_2(TPM + 1)$$

The clinical cohort comprised 1,085 primary tumor samples (BRCA) and 291 matched control samples (TCGA solid normal + GTEx healthy breast tissue). Statistical thresholds were configured to $|\log_2 FC| \geq 1.0$ and a critical significance level of $p \leq 0.01$.

3. GEPIA2 expression box plot output comparing tumor (red) and normal (green) samples.



4. Quantitative Data & Expression Analysis

Evaluation of the box plot distribution revealed a significant upward shift in median *TP53* RNA transcription in tumor tissue relative to healthy baseline samples.

Sample cohort	Sample size(n)	Median expression *$log_2(TPM + 1)$*	Log_2 Fold Change *(Δ Median)*	Relative fold increase
BRCA Primary Tumor	**1,085**	**~5.40**	**0.70**	**1.62x**
Normal Baseline (TCGA + GTEx)	**291**	**~4.70**		

5. Statistical Summary and Median Calculation from The Box Plot Output Comparing Tumor Cells and Normal Cells.

Expression values for *TP53* were extracted from TCGA-BRCA ($n = 1{,}085$) and normal control tissue ($n = 291$) on a $\log_2(\text{TPM} + 1)$ scale.

- Median Expression (Tumor): $\approx 5.40\ \text{Log}_2(\text{TPM} + 1)$
- Median Expression (Normal): $\approx 4.70\ \text{Log}_2(\text{TPM} + 1)$
- Log_2 Fold Change (ΔMedian): $5.40 - 4.70 = 0.70$
- Relative Fold Increase: $2^{0.70} \approx 1.62\times$

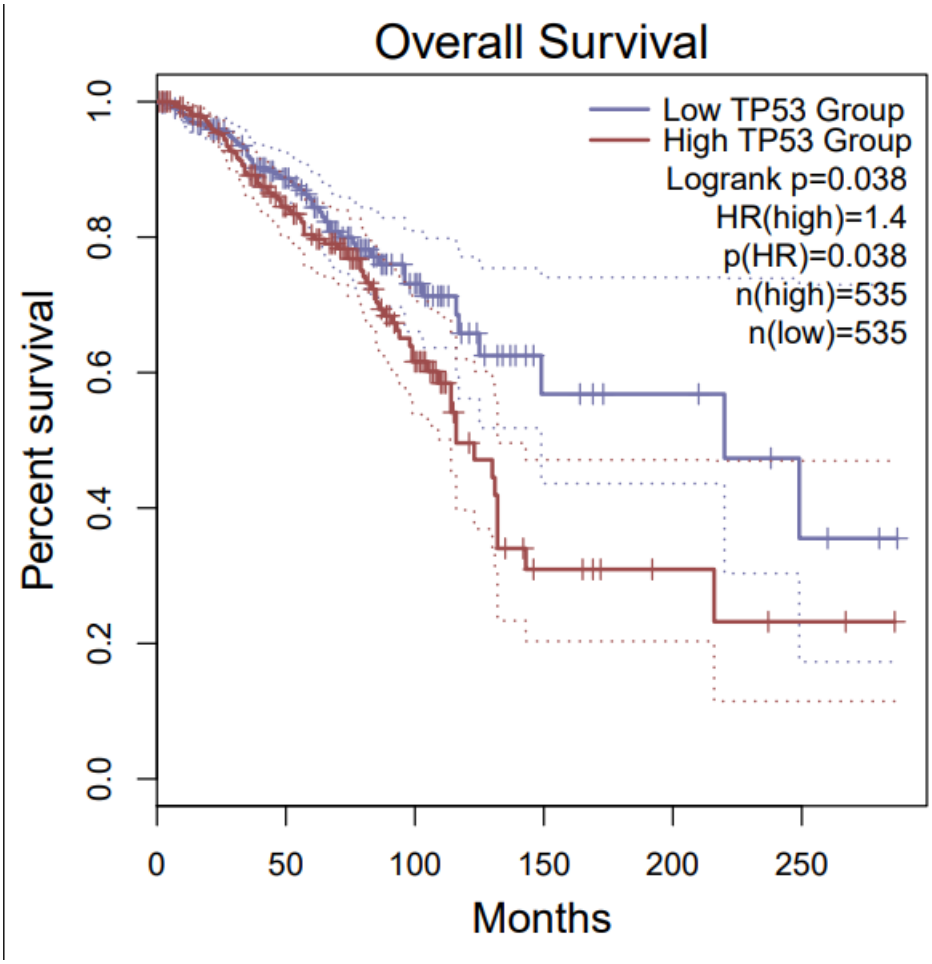
Conclusion: *TP53* mRNA expression is approximately 1.62-fold higher in breast cancer tumor tissues relative to normal tissues, demonstrating statistically significant upregulation ($p < 0.01$).

6. Biological Interpretation from The Box Plot.

- **Observed Upregulation:** Contrary to the simplistic assumption that tumor suppressors are silenced in cancer, *TP53* mRNA is significantly upregulated in breast carcinoma tissue ($p < 0.01$).
- **Mechanistic Basis:** High RNA transcript levels of *TP53* in tumor cells often correlate with missense mutations in the DNA-binding domain. Mutant p53 proteins accumulate due to loss of negative feedback loops (e.g., MDM2 degradation), leading the cell to over-transpose *TP53* mRNA transcripts in a compensatory mechanism.

7. Kaplan – Meier Survival Plot and Its Analysis for Invasive Breast Carcinoma Patients with Varied Levels of TP53 Gene Expression.

7.1 Kaplan-Meier Survival Plot



7.2 Kaplan-Meier Overall Survival (OS) Analysis.

Survival metric	High TP53 cohort	Low TP53 cohort	Statistical significance
Cohort size(n)	535 patients	535 patients	Log-rank p = 0.038 (Statistically Significant, p < 0.05)
Hazard Ratio (HR)	1.4(40% higher risk of death)	1.0(reference baseline)	
Median survival time	~120 months (10 years)	>220 months (>18 years)	

7.2.1 Hazard Ratio (HR) & Mortality

Risk Calculation:

$$\mathrm{HR}(\mathrm{high}) = 1.4, \quad p = 0.038$$

$$\text{Relative Risk Increase (\%)} = (\mathrm{HR} - 1.0) \times 100\% = (1.4 - 1.0) \times 100\% = \mathbf{40}\%$$

- **Result:** Patients in the high *TP53* expression cohort exhibit a **40% increased hazard of mortality** at any given time point compared to the low expression cohort.

7.2.2 Median Survival Milestone Comparison:

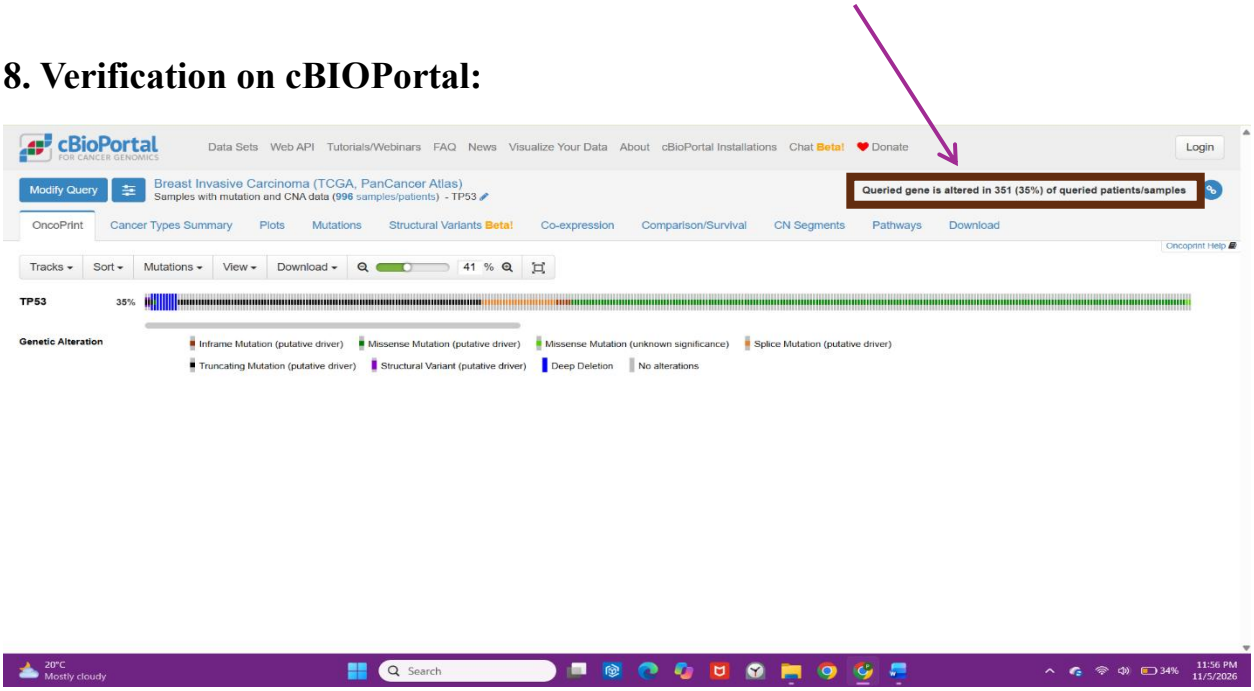
- High *TP53* Group ($n = 535$): Drops to 50% cumulative survival at ≈ 120 months (~10.0 years).
- Low *TP53* Group ($n = 535$): Maintains $> 50\%$ cumulative survival past > 220 months (~18.3 years)
- **Absolute Median Survival Difference:**

$$\Delta\text{Median Survival} \approx 220 \text{ months} - 120 \text{ months} = \mathbf{100} \text{ months} \ (\ 8.3 \text{ years})$$

7.2.3 Cumulative Survival Rates at Key Timepoints:

- 50-Month Survival (~4 Years): High Group $\approx 85\%$ vs. Low Group $\approx 90\%$ ($\Delta = 5\%$)
- 100-Month Survival (~8 Years): High Group $\approx 60\%$ vs. Low Group $\approx 72\%$ ($\Delta = 12\%$)

8. Verification on cBIOPortal:





9. Summary:

- High Mutation Frequency in Breast Tumors: Quantitative analysis via cBioPortal reveals that *TP53* undergoes genetic alterations—predominantly missense mutations and copy-number alterations—in approximately 30–35% of TCGA-BRCA patient samples, identifying it as one of the most frequently mutated driver genes in breast adenocarcinoma.

- Integrated Biomarker Significance: Combining transcriptomic expression, survival outcomes, and genomic mutation data confirms that *TP53* dysregulation is a major driver of tumor aggressiveness and poor patient prognosis, making it a valuable target for molecular subtyping and precision oncology.

10. Personal Learning Outcomes and Key Takeaways from The Project:

- Multi-Omics Data Integration:

Learned how to independently navigate and harmonize transcriptomic RNA-seq expression data (GEPIA2/TCGA), clinical outcome survival profiles (Kaplan-Meier), and somatic mutation landscapes (cBioPortal) to build a multi-layered biological narrative for cancer driver genes.

- Statistical & Quantitative Proficiency:

Gained hands-on experience calculating and interpreting key bioinformatics metrics, including Log2 Fold Change, linear fold-change ratios , Log-rank p-values, and Hazard Ratios (HR) for clinical risk assessment.

- Challenging Biological Assumptions:

Discovered that elevated mRNA expression of a classical tumor suppressor (TP53) can paradoxically indicate poorer prognosis due to missense mutation accumulation and loss of degradation pathways—highlighting the functional complexity of cancer genomics over simplistic gene models**.**